EXERCISE 6

A Second-Order Equation as a System: the Two Wronskians Coincide

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Second Order to System • Wronskian • Abel

STATEMENT

Suppose that we have a second order linear ODE $x''(t) + p(t)x'(t) + q(t)x(t) = 0$, and two solutions $x_1(t)$ and $x_2(t)$. Write it as a linear system of two first order ODEs. Check that if

$$\begin{cases} x = x_1(t) \\ y = y_1(t) \end{cases} \quad \text{and} \quad \begin{cases} x = x_2(t) \\ y = y_2(t) \end{cases}$$

are the corresponding solutions of the system, then the Wronskian as defined in this chapter coincides with the definition that we had before for second order ODEs.

PHASE 1: SETUP

p, q continuous on an interval I. Two definitions to compare: the scalar Wronskian $W_2[x_1, x_2] = x_1 x_2' - x_2 x_1'$ (Topic 3 Sec. 2.1) and the Wronskian of vector solutions $W_{sys} = \det[\mathbf{x}_1\ \mathbf{x}_2]$ (Topic 4 Sec. 3).

PHASE 2: FORMULATION: THE SYSTEM

Introduce $y = x'$ (Robinson Sec. 25.1: $x_1 \to x$, $x_2 \to x'$). Then $x' = y$ and, from the ODE, $y' = x'' = -p\,x' - q\,x = -q\,x - p\,y$:

$$\begin{cases} x' = 0 \cdot x + 1 \cdot y \\ y' = -q(t)\,x - p(t)\,y \end{cases} \qquad \dot{\mathbf{x}} = \begin{pmatrix} 0 & 1 \\ -q & -p \end{pmatrix} \mathbf{x}$$

This has the form (1) with $a_1 = 0, b_1 = 1, a_2 = -q, b_2 = -p$ (continuous). A solution $x_j(t)$ of the ODE gives the solution $(x_j, y_j) = (x_j, x_j')$ of the system, and conversely the first component of a system solution solves the ODE.

PHASE 3: DERIVATION

The corresponding solutions of the system are $\mathbf{x}_1 = (x_1, x_1')^T$ and $\mathbf{x}_2 = (x_2, x_2')^T$, so $y_j = x_j'$. Their system Wronskian is

$$W_{sys} = \begin{vmatrix} x_1 & x_2 \\ y_1 & y_2 \end{vmatrix} = x_1 y_2 - x_2 y_1 = x_1 x_2' - x_2 x_1' = W_2[x_1, x_2]$$

The two definitions coincide identically. ■

Consistency with Exercise 3

Exercise 3 applied to this system ($a_1 + b_2 = 0 + (-p) = -p$) gives $\dot{W} = -pW$, i.e. $W(t) = W(t_0)\exp\left(-\int_{t_0}^{t} p\right)$, which is Abel's identity of Topic 3 (Robinson Exercise 11.2). Direct check from the scalar definition: $W_2' = x_1'x_2' + x_1x_2'' - x_2'x_1' - x_2x_1'' = x_1x_2'' - x_2x_1'' = x_1(-px_2' - qx_2) - x_2(-px_1' - qx_1) = -p(x_1x_2' - x_2x_1') = -pW_2$, the same equation.

PHASE 4: INTERPRETATION AND CHECKS

- Therefore all the theory of systems (non-vanishing Wronskian $\Leftrightarrow$ independence, general solution, dichotomy) contains the scalar results of Topic 3 as the special case $A = \begin{pmatrix} 0 & 1 \\ -q & -p \end{pmatrix}$; the notation $\mathbf{x}(t_0) = (x(t_0), \dot{x}(t_0))$ for the initial data is the same pair as in Theorem 11.1.
- Example: $\ddot{x} + x = 0$, $x_1 = \cos t$, $x_2 = \sin t$: $W_2 = \cos^2 t + \sin^2 t = 1$ and $W_{sys} = \begin{vmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{vmatrix} = 1$; $p = 0$ so W is constant, as Abel/Liouville predict.
- The matrix $\begin{pmatrix} 0 & 1 \\ -q & -p \end{pmatrix}$ has trace $-p$ and determinant q; for constants it gives the characteristic polynomial $\mu^2 + p\mu + q$ of Topic 3.